Virtual Reality in PTSD Treatment

EST 440 – Interdisciplinary Research Methods

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1. Introduction

Advances in artificial intelligence and immersive technologies have created powerful new possibilities for healthcare innovation. Yet, within this rapid technological expansion, much of today's development is increasingly driven by investment incentives rather than meaningful social impact. Our team therefore chose to pursue a project grounded in a contrasting philosophy that technology should be designed first and foremost to improve human well-being. Among the many populations who could benefit from technology-enabled care, military personnel and veterans are among the most underserved. Post-traumatic stress disorder (PTSD) remains a significant barrier to military readiness, reintegration, and long-term mental health, yet access to effective, scalable, and safe therapeutic options continues to lag behind clinical need.

Virtual Reality (VR) offers a promising avenue for addressing this gap by enabling controlled, immersive exposure environments that align closely with the mechanisms of trauma-focused treatment. Unlike traditional exposure therapy, which often faces logistical or safety barriers, VR can deliver repeatable, customizable, and clinically supervised simulations that activate trauma-relevant cues while maintaining patient safety. Motivated by this potential, our team developed a prototype VR-based intervention intended to support PTSD treatment within a structured clinical framework. The prototype includes staged exposure environments and biometric-informed monitoring, allowing us to explore both the technical feasibility and therapeutic implications of immersive digital interventions.

The goal of this research is therefore twofold: (1) to examine VR as a scientifically grounded and clinically responsible medium for delivering trauma-focused therapy, and (2) to articulate the Scope, Technical justification, Innovation value, and Policy considerations necessary to translate such a system into a viable digital therapeutic. The following section (SCOPE) defines the population, clinical role, delivery boundaries, and operational constraints within which this VR intervention is designed to function, establishing the foundation for the technical and policy analysis that follows.

2. Scope of the Proposed VR PTSD Intervention

2.1 Defining the Scope

The scope of this intervention outlines the boundaries within which the proposed VR system operates as a therapeutic adjunct for post-traumatic stress disorder. In research design, scope refers to the clearly defined population, setting, intervention parameters, and expected outcomes that constrain and focus the study (Creswell & Creswell, 2018). Accordingly, the present scope statement specifies who the system is designed for, what clinical role it fulfills, where and how it is implemented, what boundaries limit its use, and which outcomes it intends to measure.

VR exposure therapy has shown considerable promise in facilitating extinction learning through an immersive, controllable trauma-cue environment (Kothgassner et al., 2019), with meta-analytic evidence supporting its efficacy across anxiety-related disorders (Carl et al., 2019). Rather than functioning as a replacement for trauma-focused psychotherapies or as a subordinate component within them, VR represents a broadened treatment-delivery option enabled by advancements in immersive technology (Schnurr et al., 2024; American Psychological Association, 2017).

This scope definition therefore situates the proposed VR intervention within ethical, clinical, and operational boundaries consistent with current evidence-based standards for PTSD care.

2.2 Target Population

This intervention targets combat-related PTSD among adults with confirmed diagnoses according to the Clinician-Administered PTSD Scale for DSM-5 (CAPS-5). The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), published by the American Psychiatric Association (2013), is the prevailing diagnostic framework that defines PTSD through four symptom clusters such as intrusion, avoidance, negative alterations in cognition and mood, and hyperarousal. By grounding the diagnostic threshold in DSM-5 nosology, the study maintains standardization and comparability with existing clinical research (Weathers et al., 2017).

Eligible participants include discharged veterans, active-duty returnees, peacekeeping forces (UN or NATO missions), and police or paramilitary personnel exposed to combat-equivalent trauma. These populations are prioritized because their symptom structure, dominated by cue-triggered hyperarousal, avoidance, and intrusive re-experiencing, aligns with the mechanistic affordance of VR exposure (Reger et al., 2011). Individuals with acute suicidal risk, active psychosis, or uncontrolled neurological conditions are excluded, given the ethical nd safety considerations of immersive trauma exposure (American Psychiatric Association, 2017)

2.3 Clinical Role and Boundaries

The VR system functions as an alternative delivery modality through which established trauma-focused therapies such as Prolonged Exposure (PE) and Cognitive Behavioral Therapy (CBT) can be implemented (Foa et al., 2007).

Current trauma-treatment guidelines describe VR-assisted exposure as one of several technology-enabled delivery formats for implementing evidence-based trauma therapies, reflecting the broader diversification of treatment options. The VA/DoD guideline categorizes VR exposure as an optional enhancement within existing exposure-based treatments (Schnurr et al., 2024). The APA guideline similarly frames VR within a broader spectrum of technology-assisted exposure variants (American Psychological Association, 2017).

The therapeutic objectives of the VR module are therefore reducing flashback intensity, decreasing avoidance behaviors, and lowering relapse probability, rather than achieving complete remission. This bounded clinical role maintains alignment with evidence-based standards and ensures ethical responsibility.

2.4 Delivery Environment

The intervention is implemented exclusively in controlled clinical environments, including military hospitals, veterans' medical centers, trauma-focused outpatient clinics, and behavioral health units overseen by licensed clinicians. These environments enable real-time monitoring of physiological and psychological responses, ensuring that safety overrides and clinical intervention are immediately available (Kothgassner et al., 2019).

The initial deployment stage requires full in-person supervision. A subsequent tele-supervised extension phase may be introduced once foundational safety and adherence benchmarks have been validated. This phased rollout reflects trauma-informed care principles and prevents premature unsupervised use (TIP 57: Trauma-Informed Care in Behavioral Health Services, 2014).

2.5 Intervention Modality

The VR system provides clinician-supervised, parameter-controlled VR trauma-cue exposure. Sessions simulate combat-related environments such as convoy routes, marketplaces, or urban ambush sites in immersive 3D to reproduce contextual triggers associated with PTSD symptoms (Kothgassner et al., 2019).

Clinicians regulate cue intensity, proximity, duration, pacing, and environmental variability, ensuring exposure remains within the therapeutic window. This modulation aligns with neuroscientific models of optimal arousal for extinction learning (LeDoux & Pine, 2016). The modality integrates directly with existing PE and CBT frameworks, preserving therapist oversight while adding sensory precision and repeatability (Carl et al., 2019).

2.6 Exclusion Scope

The intervention excludes all unsupervised, entertainment-based, or self-administered VR applications. Independent exposure without clinical monitoring risks symptom exacerbation, dissociation, or re-traumatization (Ehlers & Clark, 2000). Consumer VR products are not included within this scope unless they undergo medical certification and are embedded within a supervised clinical protocol.

By setting these exclusions, the system remains a professional medical intervention rather than a general wellness tool. This boundary also maintains compliance with regulatory expectations for software classified as medical devices (Center for Devices and Radiological Health, 2023).

2.7 Conclusion

Taken together, the defined population, clinical positioning, delivery constraints, intervention modality, and exclusion criteria establish a clear and ethically bounded operational framework for the proposed VR-assisted PTSD intervention. These boundaries specify not only whom the system is intended to serve but also how and under what conditions it can be safely deployed. By constraining the intervention to supervised clinical environments, integrating it as an adjunct to evidence-based trauma therapies, and delineating measurable clinical outcomes, the scope ensures methodological coherence and protects against therapeutic overreach or misuse.

This structured delineation positions the VR system as a precise, clinically responsible extension of established trauma-focused treatments, rather than a standalone or substitutive therapy. The following section (T1) therefore builds upon this foundation by examining the neuroscientific, mechanistic, and psychotherapeutic principles that justify VR as an appropriate and theoretically grounded medium for delivering trauma-focused exposure in PTSD.

3.Technology

3.1 Mechanistic Basis for Using AR/VR in PTSD Treatment

Extinction as Inhibitory Learning in Fear Circuits

PTSD is characterized not by persistent memory strength but by a failure of inhibitory control over threat memories, such that amygdala-driven threat expression is insufficiently suppressed by prefrontal regulatory systems under context supplied by the hippocampus (Milad & Quirk, 2012).

Extinction-based therapies do not erase traumatic memory traces, but they induce new inhibitory learning that competes with threat memory at retrieval and is supported by vmPFC–amygdala–hippocampus circuitry (Quirk & Mueller, 2008). Thus, any therapeutic delivery system that can repeatedly present trauma-relevant cues under objectively safe conditions while preserving context and controllable intensity is, in principle, architecturally aligned with the known mechanism by which pathological trauma memories become deactivated at retrieval (Craske et al., 2014).

Why Presence and Controllability Matter for Extinction

Extinction learning requires not only contact with the conditioned stimulus but contact under conditions encoded by the brain as "safe, predictable, and controllable" (Craske et al., 2014). VR permits controllability of intensity, proximity, temporal pacing, and sensory channels without compromising the realism of cues. This matters because if exposure is too intense, it reconsolidates fear, and if it is too weak or abstract, it fails to engage the threat system strongly enough to drive inhibitory learning (Mineka & Thomas, 1999). Conventional imaginal exposure depends on voluntary generation of imagery and therefore cannot enforce a learning environment with fixed safety contingencies, whereas VR can hold the organism inside a state space where threat cues are fully present while catastrophic outcomes never materialize (Rizzo et al., 2019). The pairing of 'treat signal present' and 'harm absent' is the exact signature required to induce a shift from threat expectancy to safety expectancy in the fear circuit (Milad & Quirk, 2012).

Context-Specificity and Variability as Requirements of Durable Change

Extinction does not automatically generalize. It is subject to renewal when context shifts, which is a signature problem in PTSD where symptoms re-emerge outside therapy rooms (Bouton, 2004). VR uniquely allows both *context preservation* (when fidelity is needed) and *context diversification* (when generalization is needed) within the same intervention architecture (Craske et al., 2014). Context variability can be instrumented by parametric manipulation of environmental geometry, soundscape density, temporal jitter, or agent behavior, without re-engineering the treatment structure. This property is non-trivial: durability of extinction depends on variability and not mere repetition, and VR is the only delivery substrate that can introduce systematic variability without changing the cognitive task or introducing real-world danger (Rizzo et al., 2019).

3.2 Technical Affordances That Map onto Mechanistic Requirements

Parametric Exposure and On-Demand Safety Gating

VR environments permit graduated threat exposure in scalar increments (e.g., distance to stimuli, number of adversarial units, sound amplitude, movement velocity) while embedding a 'hard stop' safety override under clinician control. This directly satisfies the inhibitory learning requirement that exposure be strong enough to update the generative model of threat but never escape safety controllability (Craske et al., 2014). Imaginal exposure has no equivalent safety gate because the threat is internal and not parametrically manipulable.

Multisensory Realism without Real-World Risk

PTSD triggers are rarely unimodal. They are composite textures of visual geometry, acoustic signature, temporal unfolding, spatial configuration, and proprioceptive expectation. VR allows those features to be recomposed with sufficient perceptual precision to engage the threat system without reproducing actual danger (Rizzo et al., 2019). This is the technical resolution of the ethical contradiction of trauma care: one must present the threat fully and simultaneously guarantee it is not real.

Repeatability, Standardization, and Auditability

Unlike in-vivo exposure, VR environments can be reproduced with pixel-identical fidelity across sessions, patients, and sites, allowing exposure doses to be standardized and audited. This matters because extinction is a dose-dependent learning process, where the difference between success and failure is often in dose control, not intent. VR converts a historically artisanal intervention into a parametrically governed one.

Compatibility with Closed-Loop Control using Physiological Signals

VR can be coupled to heart rate, skin conductance, pupil dilation, or respiration to maintain arousal in the optimal therapeutic band. Extinction is an arousal-contingent learning process. If arousal collapses, the session becomes supportive counseling rather than extinction, and if arousal exceeds the window, reconsolidation dominates (LeDoux & Pine, 2016). VR makes closed-loop exposure technically feasible without changing the therapeutic principle.

3.3 Alignment with the Symptom Architecture of PTSD

PTSD is a Cue-Triggered Disorder

PTSD is not a free-floating state. It is produced by context-rich, multisensory cues that are often spatially organized (Brewin et al., 2010). A delivery system that fails to reproduce the structure of those cues is mismatched to the disorder's operating architecture. VR does not treat PTSD by introducing a new mechanism but by matching the geometry of the disorder.

Avoidance is the Maintaining Factor, and VR Makes Avoidance Impossible

The strongest maintaining factor in PTSD is not memory strength but avoidance (Ehlers & Clark, 2000). Imaginal exposure is avoidable by mere refusal, but in vivo exposure is avoidable by logistics. VR removes voluntary avoidance by enclosing the patient in a perceptual field where cues are present, without escape or harm.

Reconsolidation Risk is Lower Under Programmable Safety

Uncontrolled re-exposure can reconsolidate trauma memory if safety expectancy is violated during exposure (Schiller et al., 2010). VR allows surgical control over contingencies such that expectancy violation is directed at threat predictions, not safety predictions.

3.4 Implementation and Technical Components

The prototype aims to deliver adaptive digital mental therapy for Post Traumatic Stress Disorder (PTSD) through immersive technologies with integrating VR/AR environments. This section explains the implementation ready technical components that enable real-time monitoring, personalization, and safe therapeutic progression.

Core Technical Components

VR Engine

VR engine is the foundation of the platform is a stage-based adaptive therapy pipeline built on robust, commercial grade engines such as Unity and Unreal. This architecture structures the therapeutic journey into progressive, goal-oriented modules, moving from controlled exposure to relaxation, while providing the flexibility to adapt to patient needs.

Trigger Control API

A specialized Trigger Control API provides detailed instructions for the treatment environment. This allows precise real-time modulation of certain environmental stimuli, including sound, light, motion, tactile feedback like vibration, and proximity gradient. This allows the system to dynamically increase or decrease the intensity of exposure depending on the treatment objectives.

AR Mode (Mobile / Web XR)

To bridge the gap between clinical simulations and real-world applications, AR mode is available via mobile and webXR. This feature overlays simulations that are not threatening to the patient's real-world environment, facilitating safe on-the-job training and generalization of out-of-clinical coping skills (Javanbakht et al., 2024).

Biometrics Integration

This system has powerful biometrics integration through APIs for various sensors. These include Consumer Wearables such as Apple watch, Polar and Research Devices like Open BCIs, primarily focused on capturing heart rate variability (HRV) and other key physiological pain or arousal indicators (Rubio-López et al., 2025).

Cloud Therapy OS

Cloud Therapy OS, the native backend of the cloud, serves as the central nervous system for data management and clinical supervision. Key features include detailed session recording, progress based stage unlocking, override capabilities of remote clinical practice, and comprehensive real-world evidence (RWE) logging for tracking outcomes.

A secure, cloud-native backend, the Cloud Therapy OS, serves as the central nervous system for data management and clinical oversight. Its key functions include detailed session recording, progress-based stage unlocking, and remote clinician override capabilities.

A core feature is its comprehensive Real-World Evidence (RWE) logging. This function is deliberately designed to capture data across the entire product lifecycle, aligning with modern frameworks for digital therapeutics (DTx) development, testing, and monitoring (Kim et al., 2024). This systematic collection of RWE is also essential to meet regulatory expectations, as it provides the necessary data to support regulatory decision-making for software-based medical devices (U.S. Food and Drug Administration, 2017).

Adaptive Algorithm

The key intelligence of adaptive algorithms platforms is adaptive algorithms. The algorithm synthesizes patients' subjective data such as distress self-report along with objective biometrics streamed from the integrated device. This fusion of data inputs facilitates dynamic phase transitions, allowing the rate of treatment to automatically align with the patient's real-time state (Kritikos et al., 2021).

3.5 Real-World Case Studies

The construction of a digital therapeutic prototype designed for the treatment of PTSD requires more than just theoretical speculation; it requires a foundation based on clinically demonstrated evidence of feasibility, safety, and efficacy. This section of the dissertation provides a description of four empirical cases in which the clinical and therapeutic potential of VR in the treatment of mental illness has been established and validated. Each case collectively supports the central premise that the proposed prototype is not an untested idea but a refinement of mechanisms that have been scientifically verified.

Bravemind — USC ICT & US Army

Bravemind, created by the Institute for Creative Technologies at the University of Southern California (USC) in partnership with the US Army, was the first instance of Virtual Reality (VR) exposure therapy for PTSD associated with combat (Rizzo et al, 2016). It immerses veterans and active service members in improvised explosive device training and ambushes with the goal of re-exposing them to the trauma in a safe, simulated environment. The system enables therapists to fine-tune sensory inputs, such as visual, auditory, and even olfactory, to match the patient's trauma narrative, a process termed parametric customization (Rizzo et al., 2019).

Evidence from clinical studies suggests that Bravemind is effective in alleviating ptsd symptoms. Research studies on Iraq and Afghanistan veterans show that participants receiving virtual reality exposure therapy experienced larger reductions in Clinician-Administered PTSD Scale (CAPS) scores than those in control conditions receiving standard cognitive processing therapy (Rizzo et al., 2016). The principal therapeutic mechanism is based on habituation, and occurs when a person is repeatedly exposed to a stimulus, within a richly detailed, immersive, yet safe virtual reality. Their physiological arousal and anxiety responses related to the trauma are gradually reduced.

The Bravemind project established the foundational evidence that VR exposure can replicate the benefits of in vivo therapy without the ethical and logistical risks of real-world exposure. For the current prototype, its results underscore the value of trauma-specific scenario customization and adaptive intensity control, both of which ensure patient safety and maximize treatment engagement.

Psious and Amelia Virtual Care

Psious along with Amelia Virtual Care are both examples of commercial Virtual Reality (VR) therapy applications. In fact, they integrate years of research on exposure therapy into software applications. Clinicians using the two applications are able to access multiple simulations of various environments designed to treat anxiety, phobia, and adjustment disorders. Therapists are able to customize their exposure therapy sessions using stimuli including environmental setting, visual distance, and background noise.

Clinical findings confirm that anxiety and phobia in patients show improvement. A study employing the Psious platform showed significant reductions in scores after 6-8 sessions in standardized anxiety and panic measures. This confirmed that VR-based exposure can, in fact, be as effective as traditional forms of therapy (Garcia-Palacios et al. 2021). Similarly, Amelia Virtual Care reported improvements in generalized anxiety and adaptation disorders through randomized pilot trials (Orr et al., 2023). In several European health systems, Amelia has even achieved partial insurance reimbursement, marking it as one of the few VR-based digital therapeutics recognized as a reimbursable medical device.

The success of these commercial platforms demonstrates that VR-based digital therapeutics are not confined to research laboratories but can be deployed at scale within clinical practice. This suggests that our prototype can integrate commercially validated UX frameworks, such as therapist dashboards, outcome tracking, and cloud-based data feedback, to enhance clinical usability and long-term adoption.

Oxford VR — Fear of Heights Trial

One of the University of Oxford spin-outs, Oxford VR, created a completely automated VR cognitive behavioral therapy (CBT) module for the treatment of acrophobia, a fear of heights (Freeman et al., 2018). This treatment did not involve a live therapist but an interactive virtual coach who guided users through gradual height exposure.

In a single-blind, parallel-group randomized controlled trial involving 100 subjects with a marked fear of heights, study participants who received the VR intervention averaged a 68% reduction in fear as measured by the Heights Interpretation Questionnaire (HIQ) in 5 to 6 intervention sessions (Freeman et al., 2018). At the one-month follow-up, the reduction in fear remained unchanged, demonstrating the lasting therapeutic benefits.

The Oxford VR trial was pioneering in showing for the first time that independent digital immersive therapy, without any human facilitation, is able to achieve clinically significant outcomes. This finding expands the role of VRin therapy from a potential adjunct to a stand-alone treatment. For our prototype, it strengthens the relevance of the potential of semi-automated or algorithmically guided systems that preserve a role for a clinician while providing some or a lot of autonomy to the patient.

NIMH-Supported PTSD VR RCTs (VA Hospital Studies)

Numerous randomized controlled trials assessing virtual reality exposure therapy (VRET) for PTSD in veteran populations have been funded by the National Institute of Mental Health (NIMH) in conjunction with the U.S. Department of VA. Among these studies, exposure based on VR was compared with traditional cognitive behavioral therapy (CBT) and showed not only symptom reductions that were similar to those of traditional CBT, but there was also significantly higher adherence to treatment and lesser rates of drop-outs (Olsen et al, 2022).

One of the most problematic experiences in treating PTSD is dealing with drop-outs. Thus, reduction in drop-out reduction is of great importance. By modifying the nature of VR, passive recall is replaced with active engagement, which increases patient participation and involvement. This is connected to the secondary outcomes of our prototype, which are enhanced retention, compliance, and continuity of treatment.

Conclusion

Collectively, these four case studies affirm that VR-based exposure and behavioral therapies have already demonstrated clinical efficacy, scalability, and safety across both research and commercial settings. Bravemind proved that parametric customization can effectively simulate trauma contexts. Psious and Amelia showed that clinically validated VR interventions can be productized and reimbursed. Oxford VR revealed that fully automated digital CBT is capable of achieving clinically significant results. The NIMH-supported VA studies highlighted the technology's impact on adherence and treatment retention.

Our prototype builds directly upon these validated mechanisms rather than proposing an untested concept. By translating proven VR therapeutic frameworks into a modular, adaptive platform, the project embodies a clear evidence-to-product pathway. In essence, this design approach is not about inventing therapy from scratch, but about prototyping a scientifically validated mechanism into a deployable digital therapeutic solution.

4. Innovation Value

4.0 Defining the Innovation

The 'I' in the acronym STIP (Science, Technology, and Innovation Policy) represents the most important component in understanding and knowledge creation, or in problem-solving, namely the practical part. Innovation is the point where Science and Technology intersect with the economy and the social fabric. Thus, innovation is the focal point of this application. Science is all about finding new things. It describes the social value of new ideas, methods, products, or services, the practical value of solving public problems like clean energy or public health, and the commercial value.

The innovation value of using VR in PTSD treatment is defined by its multifaceted contributions across industrial, economic, and clinical domains. It represents a paradigm shift from traditional care models by leveraging technology to enhance safety, accessibility, personalization, and scalability, all while being grounded in robust clinical evidence.

4.1 Industrial Value and Application

As a non-invasive therapeutic tool, VR treatment provides an alternative to pharmacological treatments, which often carry side effects. The industrial value of virtual reality in PTSD treatment stems from its ability to transform therapeutic principles into scalable, commercially viable applications. The key advantage of VR is that it provides a "safe and controlled environment" where exposure may take place "without the risks associated with real-world exposure" (Spytska, 2024). This safety characteristic is an important industrial advantage in and of itself. Furthermore, the technology makes it possible to create realistic simulations that are not only reusable but also highly customizable. That enables an individualized approach to treatment tailored to specific patient needs (Spytska, 2024). This allows the use of methods like gradual desensitization, a process that is difficult to standardize in the real world.

This digital nature makes the therapy highly scalable. Once a scenario is developed, it is infinitely reuseable, drastically lowering the marginal cost per treatment session. This scalability directly addresses one of healthcare's biggest challenges which is accessibility. VR platforms offer improved accessibility making consistent, evidence-based care available to patients who live in remote areas or have mobility limitations (Spytska, 2024).

Industrial potential is already being realized through tangible products including well-known software products such as "Virtual Iraq" for military use and comprehensive commercial platforms like "XRHealth", which provides a wide range of therapeutic environments. The industry's shift toward particular, high need populations is further evidenced by the creation of focused initiatives for the

"Ukrainian market" (Spytska, 2024). These examples collectively illustrate an industry is evolving from specialized, one time items to flexible, needs driven treatment platforms.

4.2 Economic Rationale and Value Creation

With the industrial value centered on accessibility and scalability, the economic rationale for VR therapy is further supported by clear market demand. There is significant interest among individuals confirming a substantial potential market is already waiting for these commercial applications (Spytska, 2024). The economic value creation is particularly compelling at a systemic level, addressing the immense costs of untreated anxiety, which are estimated at $122 billion in lost productivity and $135 billion in health care annually (Ong et al., 2022).

VR therapy can immediately reduce the financial load and demand on hospital beds by facilitating successful treatment outside of conventional hospital settings (Gómez-Bergin et al., 2023) and expanding it to the comfort and privacy of patient's homes (Ong et al., 2022). The transition to home based, remote care such as virtual reality exposure therapy (VRET) has substantial systemic benefits.

From an insurer and health system perspective, this translates into a powerful argument for cost effectiveness (Gómez-Bergin et al., 2023). Virtual reality can reduce costs compared to staging physical experiences for in vivo exposure therapy IVET (Ong et al., 2022), and can be delivered by smaller teams and in less time (Gómez-Bergin et al., 2023). The potential savings are substantial. One study cited by Ong et al. (2022) found that VRET for military PTSD saves $114,000 USD per patient compared to replacement costs. Moreover, equipment costs are no longer considered a major obstacle, with some research using devices as cheap as $5 USD (Ong et al., 2022). However, systemic hurdles like investment in formal training (Ong et al., 2022) and high upfront software costs (Spytska, 2024) remain. Ultimately, the economic rationale is clear but the current challenge is leveraging this data to justify the initial high costs and unlock systemic reimbursement.

4.3 Differentiation from Conventional Healthcare Paradiagms

The core differentiator lies in its classification as a novel therapeutic modality: 'Therapy as Software' (TaaS). Unlike traditional healthcare models that require co-location of patient and provider, this model bifurcates the clinical decision from its delivery. While a qualified clinician prescribes the therapeutic protocol, the treatment itself is delivered remotely via the software platform. This "prescribed-but-remote" paradigm maintains rigorous clinical oversight while maximizing scalability, patient convenience, and data-driven insights, representing a fundamental departure from legacy care models.

Traditional exposure therapy is operationally difficult. For traumas, in vivo (real-life) exposure is either expensive, impractical, or just risky. The alternative imaginal exposure is less immersive and mostly depends on a patient's capacity for visualization, which many find challenging (Kothgassner et al., 2019). Virtual reality exposure treatment (VRET) was developed as a viable approach to overcoming these intricate problems (Kothgassner et al., 2019) by providing a safe and controlled environment (Sigal, 2024), which enables simulating environments that are difficult, unsafe, or costly.

The prescribed but remote paradigm is made possible by this TaaS model. Unlike traditional models which require co-location, a clinician can prescribe a VRET protocol that the patient can access remotely. By extending treatment to the comfort and privacy of patients' location, directly addressing the geographic and economic barriers that limit traditional therapy, especially in rural areas or for individuals with mobility issues (Sigal, 2024).

The distinction preserves clinical rigor in addition to being logistical. There was no significant difference between VRET and active comparators such as traditional therapy in reducing PTSD symptoms (Kothgassner et al., 2019). Moreover, VRET has demonstrated equal or better outcomes with lower dropout rates than traditional in-vivo exposure (Ong et al., 2022) and could prove more engaging for some individuals (Sigal, 2024). VR for PTSD is not just a new tool. It is a new modality that keeps the gold standard's effectiveness while fundamentally solving its core logistical, economic, and accessibility obstacles.

4.4 Personalization Innovation

This technology moves beyond the limitations of static, fixed-protocol treatments. By using real-time biometric data to produce a closed-loop, adaptable therapeutic experience. It represents a substantial personalization advance. The goal is to create tailored therapies that appropriately represent each patient's specific traumatic experiences, but this process needs to be handled cautiously (Van Veelen et al., 2022). In order to maintain the patient in a tolerable therapeutic state, it is crucial to be able to monitor the patient's response to the immersion objectively (Van Veelen et al., 2022).

Instead of relying solely on patient self-report, use adaptive algorithms to examine physiological signals in real time. These inputs may consist of pupil diameter measurements or eye-tracking data (Javanbakht et al., 2024). Based on the patient's instantaneous psychophysiological reactions, monitoring and altering the exposure scenarios according to this stream of biofeedback (Javanbakht et al., 2024). By guaranteeing that the intervention is constantly exactly matched to the user's situation, this adaptive approach dynamically modifies the therapeutic difficulty, maximizing patient engagement and helping to alleviate the significant dropout rates that frequently afflict traditional therapies (Javanbakht et al., 2024).

4.5 Conclusion

The VR innovation value for treating PTSD rests on the compelling intersection of these industrial, economic, and clinical paradigm shifts. It is the first move of many that shifts from a static, co-located service to a dynamic, "prescribed-but-remote" therapeutic platform (Therapy as Software). This model fulfills the industrial criterion of a scalable, low-cost, and reusable product (4.1), systemic economy criterion of a solution that alleviates the hospital burden and is cost-effective for insurers (4.2), and the clinical criterion of a treatment that is safer and more easily accessed while "gold standard" effective (4.3).

The system expectancy violation (4.4) is transformed from the corrective 'adaptive' prototypically and 'adaptive' individualized protocol inferred from the one-size-fits-all therapeutic paradigm to a responsive, closed-loop 'ecosystem' 'interventions' system. The objective mastery of 'tolerance' as her 'window of control' is a critical innovation to address the high patient dropout rates, which are targeted at solving the elephant in the room of conventional care. The case of VR for PTSD meets the core definition of innovation. It creates social value – public health improvement, industrial value - new market creation, and economic value - systemic cost reduction, technology applied for its practical use, and is therefore an innovation in mental healthcare.

5. Policy and Regulatory Considerations (P of STIP)

VR-based digital therapeutics are being developed for PTSD and other psychiatric conditions and are increasingly being recognized as valid treatment options. The development of these therapeutics will depend, in part, on relevant policy frameworks, in addition to scientific feasibility and technological advancement. In the STIP, the “P” emphasizes the regulatory, institutional, and administrative structures that determine whether an intervention can be safely adopted, ethically justified, and sustainably integrated into real clinical systems. When developing a therapy designed for high-risk populations like active-duty service members and veterans, potential policies primarily come into play when developers must address complex oversight systems, data protection, and reimbursement policies. These policy nuances become key in transforming a well-designed VR exposure therapy prototype into a clinically usable, legally compliant, and scalable digital therapeutic.

5.1 Classification Issues

A first regulatory consideration regarding VR-based PTSD digital therapeutics is how to classify the solution legally under current medical-software regulatory schemes. In the United States, mental wellness apps that provide general stress relief or meditative support are unregulated as medical devices because they do not make a clinical treatment claim, falling under the FDA's portal on “General Wellness”. On the other hand, in cases of treating specific disorders, including PTSD, the software in question must undergo the evaluation referred to as Software as a Medical Device (SaMD). This process necessitates the proof of safety, as well as therapeutic effect, prior to market entry (U.S. Food and Drug Administration, 2023). Digital Therapeutics (DTx) comprises a more strictly defined subclass of SaMD, where the therapeutic action is implemented through the software solution, rather than through observation and education. Selecting the right class will influence the extent of clinical validation needed, the regulatory route (e.g. 510(k), de novo), and opportunities for reimbursement (Lakhan et al. 2025). Considering the algorithmic exposure therapy and purported symptom alleviation, a VR intervention for PTSD would most likely be categorized as a DTx, which would imply stringent requirements for evidence production and monitoring.

5.2 Military and VA Policy Layers

For deployment in military and veteran contexts, the prototype must navigate regulatory processes distinct from those governing civilian healthcare. Before technologies are implemented or integrated into new applications, the Department of Veterans Affairs (VA), the largest integrated healthcare system in the nation, must receive and review the internal assessment and evaluation of the Veterans Health Administration’s Innovation Ecosystem and the Office of Mental Health and Suicide Prevention (U.S. Department of Veterans Affairs. n.d). Similarly, there is also a need for approval from the Department of Defense (DoD) and the Defense Health Agency concerning active-duty military uses, concentrating on the operational safety and the clinical appropriateness of the system for combat-exposed individuals, as well as the security interoperability with the military medical-records systems (Defense Health Agency, n.d.). Since PTSD treatments may involve trauma narratives linked to sensitive operations, these institutions are subject to more scrutiny than civilian hospitals. Therefore, the integration of regulations must address the efficacy of the PTSD treatments clinically, as well as the policies related to the military mental-health system.

5.3 Data Governance and Protection

Data related to PTSD constitute highly confidential personal information, particularly concerning military members, as their trauma histories may overlap with details of their deployments, mission documentation, or other sensitive information. Any information about patients that is collected through the therapeutic system is subject to the Health Insurance Portability and Accountability Act (HIPAA), which requires that the information be stored in an encrypted manner, be accessed in a manner that is regulated, and be subject to controlled flows.

In jurisdictions governed by the General Data Protection Regulation (GDPR), biometric and mental-health data qualify as “special-category” data that require enhanced legal protections and explicit consent frameworks (European Parliament, 2016). Studies indicate that military mental-health data require additional safeguards because breaches could implicate national security as well as personal privacy (Wong et al., 2025). Therefore, it has become necessary for the system to have a safe governance structure that will compartmentalize trauma-related information, restrict the unauthorized secondary use of data, and lessen the danger of the information being operational in cyberspace. Considering the above sensitivities, the system must have a safe governance structure that can compartment trauma data, restrict unauthorized secondary use, and lessen the danger of operational data leakage. One approach to the implementation of these features may be the use of edge computing, a technology that processes sensitive data locally and not in a centralized cloud. This approach has been shown to reduce the volume of data transmitted off-device, thereby lowering the risk of data breach or unauthorized exposure while enabling real-time processing and auditability in privacy-critical health applications (Nawaz et al., 2025).

5.4 Liability and Safety Protocols

The methodology of exposure-based therapy may pose risk factors such as surging levels of anxiety, dissociation of self, or withdrawal from therapy due to eruptions of overwhelming distress. Studies about VR exposure therapy show that the symptom over-response is the result of exposure that is mislabeled as being therapeutic, or from exposure unaccompanied by real-time behavioral monitoring, emphasizing the necessity of rigid control of exposure therapies to prevent symptom over-response (Maples-Keller et al., 2017). In order to reduce risk of legal liability and to protect patient safety, the model must have a fully functioning “kill switch” so that clinicians or the model can immediately stop or dilute the simulation any time distress, whether embodied, or through self-reporting, exceeds safety thresholds. Formal emergency procedures, risk-mitigation protocols, and clinician guidance documents should be fully integrated into the therapeutic workflow. Absence of these protections may lead providers to encounters of malpractice exposure or negligently unaddressed device claims if the therapy negatively impacts the condition of the patients.

5.5 Insurance and Reimbursement Pathways

To ensure meaningful implementation in clinical settings, the digital therapeutic must be supported with a reimbursement strategy. In the U.S., digital approaches, including those delivered through VR, should have insurable coverage by existing medical billing codes, for instance, programs can be billed using CPT or HCPCS codes as genuine medical services. In fact, as of October 2023, the HCPCS Level II code E1905 has been specifically created for “virtual reality cognitive behavioral therapy device (CBT), including pre-programmed therapy software,” thus incorporating VR therapy into the Durable Medical Equipment (DME) benefit category (HCPCS Codes, n.d.). This situation indicates that once a system has obtained regulatory clearance by the FDA, it may also have the opportunity to obtain formal reimbursement.

Value-based contracting might represent yet another alternative beyond traditional fee-for-service billing whereby payers might be willing to reimburse certain treatments when measurable clinical or economic outcomes have been achieved. For example, reductions in symptoms of PTSD, improved adherence, lowered dropout rates, or long-term savings from decreased comorbidities (Reger et al., 2016). Such outcome-based models dovetail with ongoing initiatives for value-based care and, consequently, digital therapeutics might become even more attractive to payers and institutions when long-term benefits are documented.

An important aspect of being able to reimburse clients is obtaining FDA clearance. Without FDA clearance, a device or software is unable to formally obtain reimbursement, as only FDA cleared software can receive CPT and HCPCS codes (U.S. Food and Drug Administration, n.d.). Once a device is cleared, the therapy is billable as any other medical service within the Department of Veterans Affairs (VA) and other systems of military health care.

A real-world example illustrating a viable reimbursement pathway is AppliedVR, a digital therapeutic company that developed an FDA-cleared virtual reality program for chronic lower back pain. Their product, RelieVRx, became the first VR-based therapeutic device to receive FDA De Novo clearance for pain management and has since secured contracts with the U.S. Department of VA, enabling veterans to access the therapy through clinician prescriptions and allowing the device to be billed similarly to standard medical care within VA and military healthcare systems (Lovett, 2023). This case demonstrates that, once a VR therapeutic obtains regulatory clearance and appropriate billing codes, integration into federal healthcare systems becomes possible.

A real-world instance demonstrating a potential reimbursement pathway is AppliedVR. They offer a digital therapeutic that features a virtual reality program to treat chronic lower back pain. Their product represents the first VR-based pain management therapeutic to achieve FDA De Novo clearance. The company has contracts with the U.S. Department of VA, which permits veterans to receive the therapy and prescriptions from clinicians, and allows the device to be billed as traditional care within the VA and military healthcare systems (Lovett, 2023). This represents the first instance of a VR therapeutic program that has been incorporated into healthcare systems, which increases the potential for other digital therapeutic programs especially for VR as well.

Obtaining FDA clearance is critical for any therapeutic to obtain. The fee schedule for public submissions shows that the cost for a 510(k) submission is between $6,084 to $24,335, and with external consulting, testing, and registration, the costs can reach well over $30,000 for even the most basic submissions (510kFDA, n.d.). The costs for devices with no predicate devices and needing a De Novo pathway are even more expensive.

To meet these financial and regulatory demands, we propose an integrated strategy grounded in established digital-therapeutic commercialization models. Building alignment between system design and FDA requirements early on, such as employing predefined risk-mitigation controls, usability testing, cybersecurity controls, and clinical validation endpoints, tends to minimize time-consuming and expensive revisions and optimizes the approval process. Support from the Federal Government through research mechanisms such as the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs help mitigate costs associated with early development and regulatory processes. Furthermore, partnerships with academic medical centers and VA-affiliated research hospitals are beneficial for conducting clinical trials at low costs and obtaining the robust real-world evidence needed for FDA clearance and reimbursement subsequent to FDA clearance (National Institutes of Health, n.d.). Finally, After receiving FDA clearance and obtaining CPT and HCPCS codes like AppliedVR's HCPCS Level II designation, creates a foundation for establishing partnerships with the VA and Military healthcare systems, ultimately allowing PTSD-focused VR exposure therapy to be billed as standard reimbursable care.

Through this sequence, which is insurance framework identification, regulatory clearance, real-world evidence generation, code assignment, and VA integration, our PTSD-targeted VRET can follow a proven reimbursement model while distributing regulatory costs across grant funding, academic partnerships, and federally subsidized research mechanisms. This approach provides a practical and financially sustainable pathway toward national deployment within military and veteran healthcare infrastructures.

5.6 Conclusion

The need for a multifaceted policy approach is evident from the regulatory, institutional, and reimbursement aspects surrounding a VR-based PTSD digital therapeutic. Classifying it as a DTx-grade SaMD provides a framework for generating clinical evidence, while military and VA-specific stipulations address the particular needs of trauma-exposed service members. Data governance safeguards provide privacy and national security, and the safety protocols balance clinical and legal protections. Also, establishing a viable reimbursement pathway is necessary for sustainability and seamless integration into standard practice. Considering these factors early in the development process not only streamlines building regulatory readiness, but also enhances the chances of the prototype progressing into a thoroughly vetted, ethically deployable, and easily available therapeutic for PTSD.

6. Prototype Implementation

For this project, we built an early-stage VR therapy prototype that allows users to experience controlled exposure in a gradual and safe manner. The system combines hand-tracking interactions with scene-based stimuli, and each stage ends with a short self-reflection prompt so the user can report their anxiety level to a specialist.

6.1 System Overview

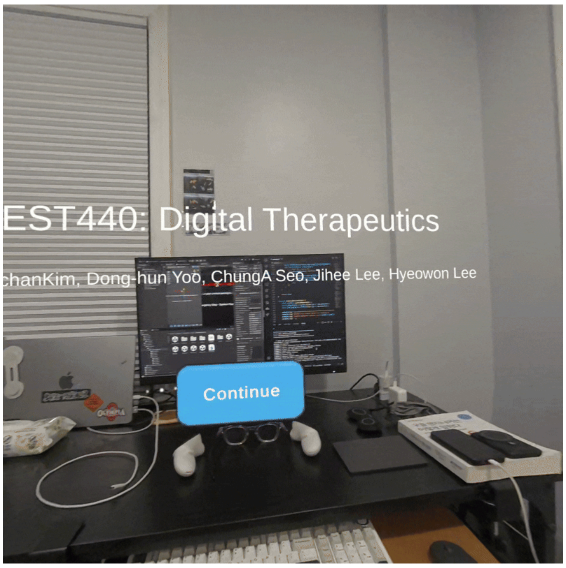
The prototype was developed using Unity 6 and deployed on the Meta Quest platform. The device's hand-tracking capabilities play a key role: both hands are captured every frame through 26 joint anchors each, which allows us to interpret simple gestures without relying on controllers. To move from one scene to the next, the user simply claps, a gesture chosen because it is intuitive and does not disrupt immersion. The full session consists of three exposure stages:

- Sound Exposure
- Audio–Visual Exposure
- Field Hospital Exposure

Each stage focuses on a specific type of trigger and gradually increases the level of sensory input.

6.2 Stage 1: Sound Exposure

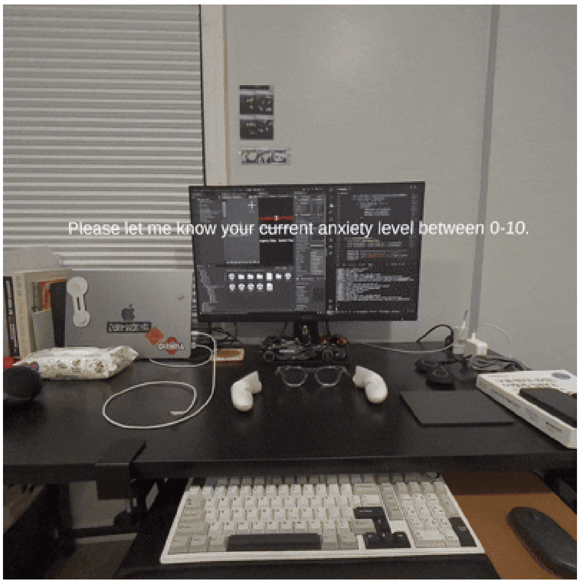
The first stage focuses entirely on audio. The user hears sounds such as gunshots or explosions, which are common triggers for individuals with combat-related trauma. Since no visual elements are added here, we can observe how the user reacts to noise alone, without additional context. When the scene ends, a short anxiety question appears in VR, and the answer is later discussed with a specialist.



Intro → Sound-Only Stage Transition

6.3 Stage 2: Audio–Visual Exposure

In the second stage, the real world slowly fades out and a virtual forest begins to appear around the user. Along with rustling leaves and distant voices, more intense sounds such as gunfire are introduced. Trees, mountains, and grass fill the user's field of view. This combination helps us examine how the user responds when sound and visuals are presented together. As before, the scene concludes with a brief anxiety-level question.

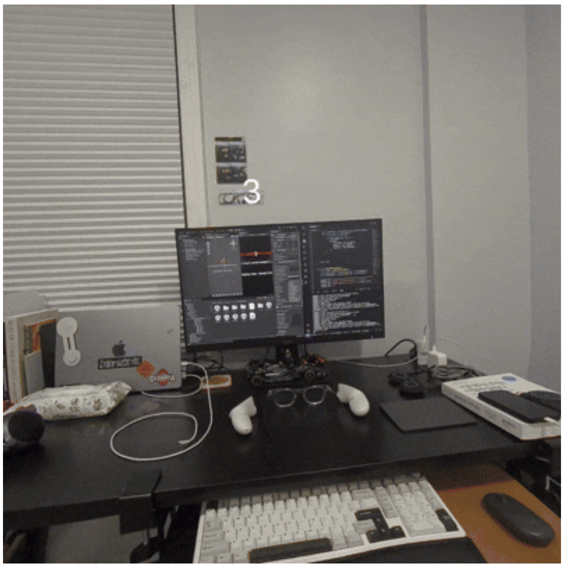


Real World → Virtual Reality Fading out

Anxiety Questions on Real World

6.4 Stage 3: Field Hospital Scene

The final stage recreates a military field hospital. The setting includes injured soldiers, medical equipment, and pain-related audio cues. Although hospitals may seem familiar in everyday life, the combat-related context can be overwhelming for individuals with PTSD. Because of its emotional weight, this stage is placed last. A final self-assessment question marks the end of the therapy session.

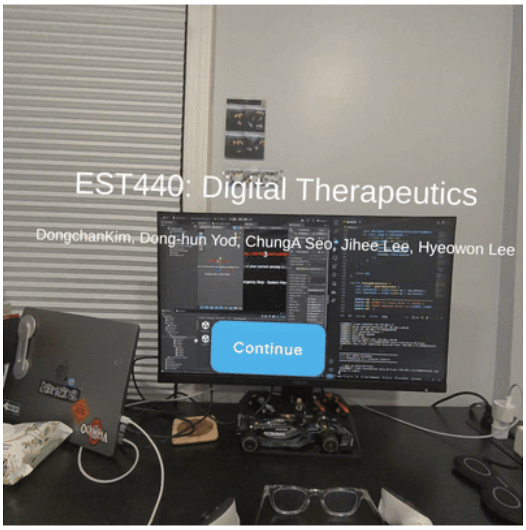


Real World → Virtual Reality Fading out

Anxiety Questions on Real World

6.5 Kill Switch Feature

To ensure safety, the system includes a remote Kill Switch. If the specialist monitoring the session notices signs of distress, they can stop the VR experience immediately without waiting for user input. This feature is supported by a small Node.js server that uses WebSocket communication to send real-time commands to the device.



Kill Switch Activation

6.6 Technology Stack Summary

Core Platform

- Unity 6 (6000.0.45f1)
- Meta Quest VR/AR
- Universal Render Pipeline (URP) 17.0.4

XR/VR Packages

- Meta XR SDK 78.0.0
- XR Interaction Toolkit 3.0.9
- XR Hands 1.5.1
- OpenXR 1.14.3

Input System

- Unity Input System 1.14.0

Key Features

- Full Hand Tracking
- Spatial Anchors
- Scene API
- Render Model
- WebSocket-based remote Kill Switch (Node.js)

6.7 Cost / resource

Three types of resources are required for implementing a VR-based PTSD therapeutic system: hardware, software, and content creation. Clinically deployable virtual reality devices, such as the Meta Quest series, are comparatively more affordable than traditional medical devices, as they only cost a few hundred dollars. Most clinics tend to have several units to allow for cleaning between patients, but the total amount of hardware needed by clinics is small. Additional computer resources that are needed for scene rendering, data encryption, and for the clinician dashboards can be satisfied by ordinary workstations and currently available cloud services.

Therapeutic environments have to be created and maintained as a second cost layer. There are some considerable differences between the patients' trauma cues, and as a result, VR scenes tend to require lightweight personalisation, like changing environmental components, sounds, or distance to the patient. Most clinical VR software, rather than building fully unique environments from scratch, draws from modular libraries of assets, thus reducing the amount of building needed. The result of this is that the configuration cost is small, with the only larger consideration being clinician configuration time rather than expensive software development.

Lastly, there are ongoing costs associated with logging data from each session, storing data remotely, and conducting continuous monitoring. These activities generally function on the basis of a subscription or licensing model, which means that the marginal cost of taking on each additional patient decreases as the service is scaled up. Overall, the costs incurred are lower than those of traditional exposure therapies, which are far more resource-intensive in terms of staff time and physical resource allocation. Despite this, the option may offer a lower level of exposure, be more consistent, more repeatable, and more scalable in the long term.

7. Conclusion

During the advancement of this project, a prototype has been created that uniquely integrates neuroscience, clinical psychology, and immersive technologies. Because of the neurobiological principles that the prototype system implements, VR technology has become much more than just a “buzzword” and has become a legitimate therapeutic tool. Veterans may utilize this to address the traumas they have experienced themselves. Exposure of traumas, with the system, is designed to provide safety controls and virtual environments. This positions the virtual reality system as an adjunct to trauma therapy, rather than an alternative with little to no grounding.

The advanced prototype demonstrates the potential for the 'Therapy-as-Software' approach within the confines of clinical and operational safety and compliance. The system is designed with safety features governing the system ("Kill Switch) to make it adaptable for clinical environments. The system is a prototype that is beyond proof of concept. Classifying the system as Software as a Medical Device demonstrates a clear understanding of HIPAA and GDPR, and data governance. This facilitates the primary objective of the system, therapeutic rigor, while also alleviating the logistical burden that has typically limited access to more traditional models of therapeutic care.

Within a Science, Technology, and Innovation Policy (STIP) framework, the intervention operates at the nexus of the science, technology, and social good triad. Being a digital therapeutic that is reusable and has a low marginal cost, the intervention is a significant contributor to the accessibility and affordability of clinically unaltered, high-quality care. The project demonstrates that the most profound shifts in mental health care innovation originate not from technology alone, but from the optimal balance of technological potential, protective policies, empirical evidence, ethical considerations, and effective policy. The system is a clinically responsible adaptation of the gold standard in PTSD treatment that bolsters clinicians’ and extended trauma care to high-impact and underserved populations.

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